

Theory Building



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Course Content and Structure

Overview

1. What is science?
2. Scientific research
3. Qualitative data analysis

Theory building

4. **Theory building**
5. Systematic reviews
6. Qualitative surveys
7. Action research
8. Case study research

Theory validation

9. Theory validation
10. Survey research
11. Controlled experiments

Comprehensive

12. Design science

Academia

13. Academic writing
14. Academic publishing

Agenda

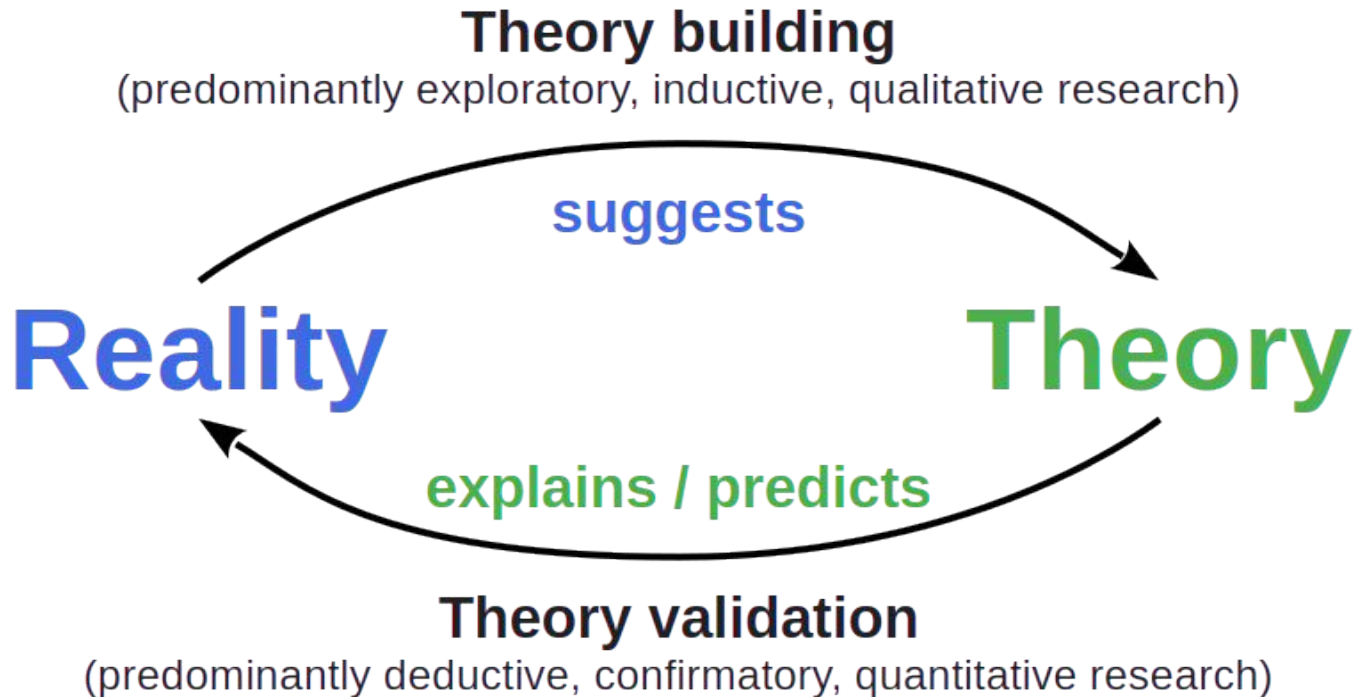
1. Theory building
2. Qualitative research
3. Example methodologies
4. Multi-method research
5. Mixed-methods research
6. Quality assurance

1. Theory Building



The Logic and Process of Science (Recap)

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Purpose of Theory Building

The purpose of scientific theory building is to

- Create, build out, and evaluate a theory that
 - Correctly explains and/or predicts reality
 - Can be continuously tested
 - In a cost-efficient way

Terminology (theory building, development, etc.) may vary

Types of Research Questions Answered

Explanatory questions like how, why?

- Questions that lead to a theory

Not: Whether something is the case

Theory Building Research

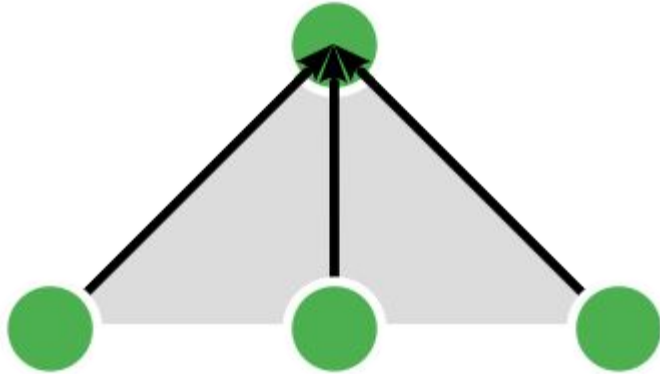
Theory building research is predominantly

- Exploratory
- Inductive
- Qualitative

Inductive Reasoning

Inductive research is based on **inductive reasoning** which is

- Abstracting from data by recognizing patterns and drawing conclusions



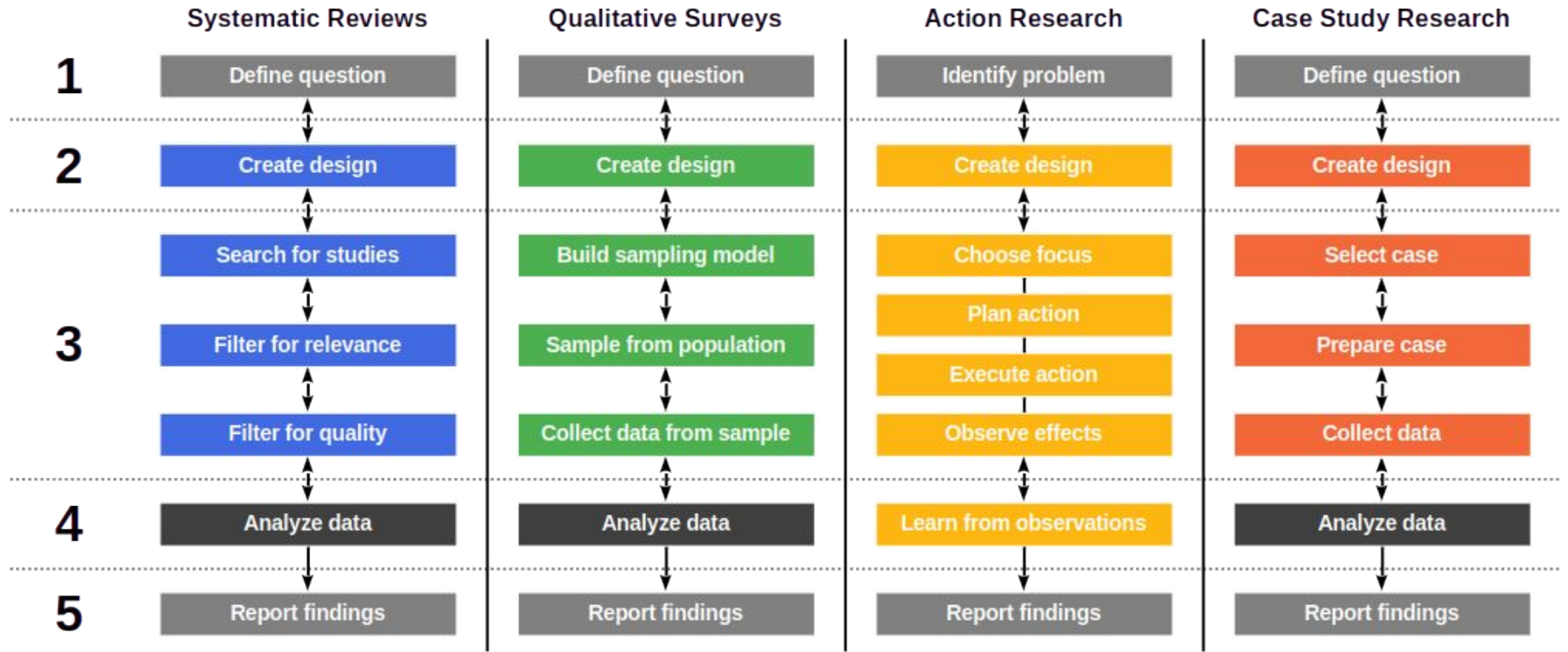
Theory Building Process



The **theory building process**

- Is iterative
- Is incremental
- Mitigates risk
- Never ends

Theory Building Processes by Methodology



What Makes a Good Theory? [1]

1. Testability
 - The degree to which empirical refutation is possible
2. Empirical support
 - The degree to which empirical studies support the theory
3. Explanatory power
 - The degree to which all known observations can be explained
4. Parsimony
 - The theory's simplicity
5. Generality
 - The theory's breadth and width
6. Utility
 - The theory's applicability

2. Qualitative Research



Base Terminology (Recap)

Researchers use a research methodology

- Recap: A start-to-finish framework for performing theory building

To create a research design

- Recap: A process description for answering a research question

That utilizes research methods

- Recap: A method answering a type of research question

Which combine research practices

- Recap: A way of doing something with a defined outcome

What Makes a Research Design a Qualitative Design?

1. The **theory building** purpose
2. The use of **purposive sampling** in data collection
3. The acquisition and use of **qualitative data**

Qualitative vs. Quantitative Research Methods (Recap)

The diagram is a 2x2 matrix. The vertical axis is labeled 'Research method' in a blue box on the left, with 'Quali-tative' at the top and 'Quanti-tative' at the bottom. The horizontal axis is labeled 'Research purpose' in a green box at the bottom, with 'Theory building' on the left and 'Theory validation' on the right. The cells contain the following text: Top-left (white background) is 'Strongly aligned'; Top-right (gray background) is 'N / A'; Bottom-left (gray background) is 'Infrequently used'; Bottom-right (white background) is 'Strongly aligned'.

Research method	Quali-tative	Strongly aligned	N / A
	Quanti-tative	Infrequently used	Strongly aligned
		Theory building	Theory validation
Research purpose			

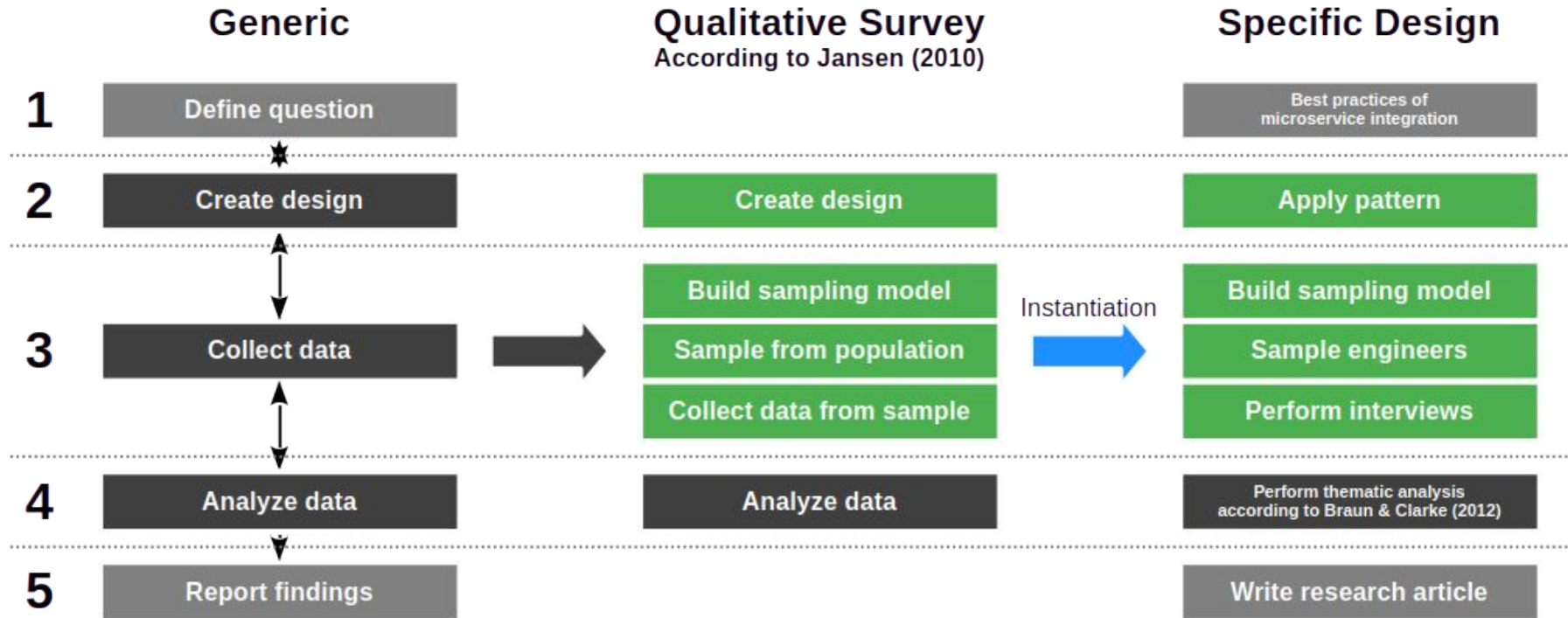
Example Qualitative Research Methodology (Recap)

The **qualitative survey** according to Jansen (2010)

1. Write research protocol
2. Build sampling model
3. Sample from population
4. Collect data from samples
5. Analyze data from samples
6. Iterate or conclude

Example Research Design Instantiation

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3. Example Methodologies



Research Methodologies

Main categories

- **Systematic review**
- **Qualitative survey [1]**
- **Action research**
- **Case study research**
- Grounded theory
- Ethnographies

Not a methodology

- Introspection

Action Research

Action research is a research methodology in which the researcher

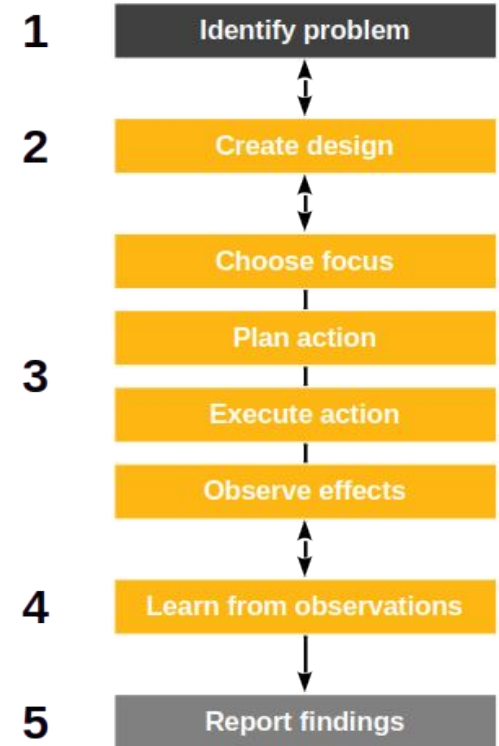
- Applies a theory being built to help create a desired outcome
- Evaluates the theory as to its trustworthiness

Participatory action research is action research in which the researcher

- Joins a case (organization) to work side-by-side with practitioners

The Action Research Process

1. Identify (research) problem
2. Create research design
3. Perform action
 - a. Choose focus
 - b. Plan action
 - c. Execute action
 - d. Observe effects
4. Learn from observations



Example Action Research Methodologies

- Lewin (1946): Action research
- McIntyre (2007): Participatory action research
- Kemmis et al. (2014): Critical participatory action research

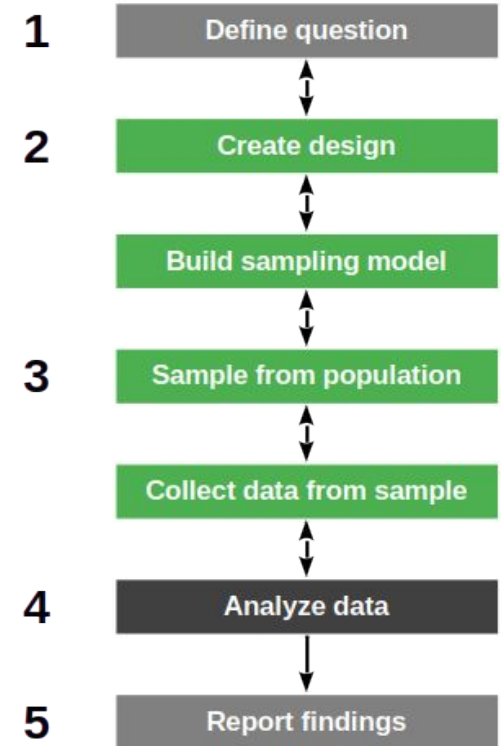
Case Study Research

Case study research is a research methodology in which the researcher

- Investigates a (contemporary) phenomenon in its original context
- Where the boundaries between phenomenon and context are blurry

Case Study Research Process

1. Define research question
2. Create research design
3. Iterate over
 - a. Select case
 - b. Prepare case
 - c. Collect data
 - d. Analyze data
4. Report findings



Example Case Study Research Methodologies

- Eisenhardt (1989): Case study research
- Yin (2009): Case study research
- Runeson et al. (2012): Case study research in software engineering

Research Paradigms

- Analytical research
 - Research is detached; goal is to understand
- Action research [1]
 - Research is involved; goal is to improve
- Design science research
 - Research is applied; goal is to innovate

4. Multi-Method Research



Multi-Method Research Designs

A **multi-method research design** is a research design that

- Uses multiple research methodologies for theory building

Such methodological triangulation improves trustworthiness

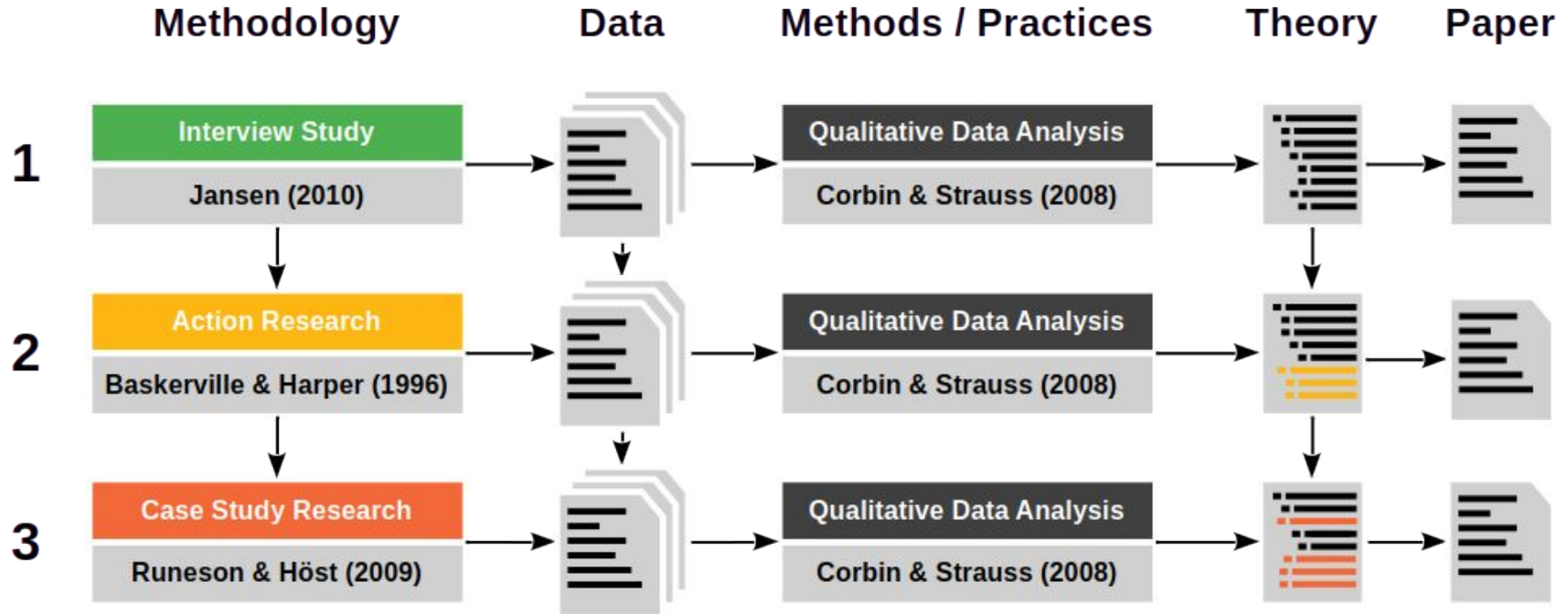
Example Multi-Method Research Design

Best practices of microservices integration

1. Theory creation
 - a. Jansen (2010): The qualitative survey
2. Theory build-out
 - a. Baskerville & Harper (1996): Participatory action research
3. Theory evaluation
 - a. Runeson & Höst (2009): Case study research

The Importance of Qualitative Data Analysis

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5. Mixed-Methods Research



Mixed-Methods Research Design

A **mixed-methods research design** is a research design that

- Combines qualitative with quantitative research methods

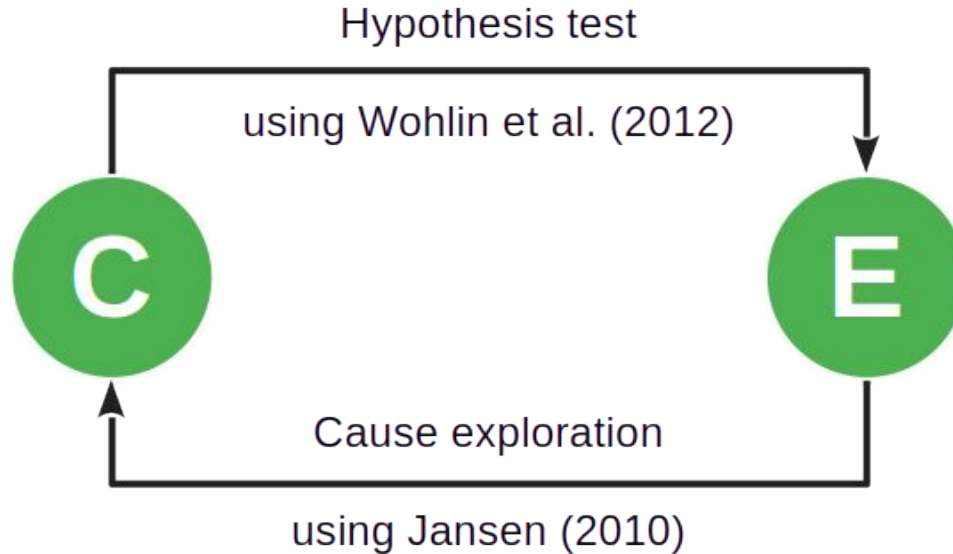
Example Mixed-Method Research Design

Novel programming language feature

1. Hypothesis creation
 - a. Jansen (2010): The qualitative survey
2. Hypothesis test
 - a. Wohlin (2012): Controlled experiment
3. Theory building
 - a. Jansen (2010): The qualitative survey

Causal Analysis / Correlation and Causation

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6. Quality Assurance

Quality Criteria for Research Methods (Recap)

Intuition	Qualitative research	Quantitative research
Truth value	Credibility	Internal validity
Applicability	Transferability	External validity
Consistency	Dependability	Reliability
Neutrality	Confirmability	Objectivity

Trustworthiness

Credibility is

- The degree of confidence in the truth of the findings

Transferability is

- The degree to which findings can be transferred to another context

Dependability is

- The degree to which findings are stable over time

Confirmability is

- The degree to which the findings can be confirmed by other researchers

Establishing Quality (Not: Assuring it)

Quality in qualitative research is “established”

- Quality is proactively baked into the methods

Practices for Establishing Quality [1]

Credibility

- Prolonged Engagement
- Persistent Observation
- (Different forms of) triangulation
- Peer debriefing
- Negative case analysis
- Referential adequacy
- Member-checking

Transferability

- Thick description

Dependability

- Inquiry audit

Confirmability

- Confirmability audit
- Audit trail
- Triangulation
- Reflexivity

Summary

1. Theory building
2. Qualitative research
3. Example methodologies
4. Multi-method research
5. Mixed-methods research
6. Quality assurance

Thank you! Any questions?



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